

Network Operations Strategy Memo

To: Head of Network Operations, Delhivery

From: Data Science & Logistics Intelligence Team

Subject: Optimizing Delivery ETAs and Hub Operations via Graph-Based Network Intelligence

Executive Summary

Delhivery's current OSRM routing engine systematically underestimates actual travel times across the network. By modeling the logistics network as a directed weighted graph, we have identified that this underestimation is not random; it is structurally concentrated around specific chokepoint hubs and chronically delayed corridors.

By integrating graph-based features (betweenness centrality and Node2Vec embeddings) into our ETA model, we reduced the Mean Absolute Error (MAE) by **22.54%** (saving 10.42 minutes of prediction error per segment) and increased prediction accuracy (predictions within 15% of actual) by **7.43% absolute** (raising baseline accuracy from 68.26% to **75.69%**, exceeding our 70% operational threshold).

Upgrading our top 3 bottleneck hubs (Gurgaon Bilaspur, Bangalore Nelamangala, and Bhiwandi Mankoli) to reduce their dwell and processing delays by 50% will prevent **1.83% of late deliveries network-wide**, recovering significant revenue-at-risk and improving SLA compliance.

Top 5 Bottleneck Hubs

Using structural betweenness centrality and historical delay minutes, we ranked the top 5 bottleneck hubs in the Delhivery network:

Rank	Hub ID	Hub Name	Betweenness Centrality	Cumulative Delay (Mins)	SLA Breach Rate
1	IND000000ACB	Gurgaon Bilaspur HB (Haryana)	0.2202	498,705	95.8%
2	IND562132AAA	Bangalore Nelamangala H (Karnataka)	0.1250	258,387	94.9%
3	IND712311AAA	Kolkata Dankuni HB (West Bengal)	0.0910	145,505	100.0%
4	IND421302AAG	Bhiwandi Mankoli HB (Maharashtra)	0.0561	221,960	99.8%
5	IND501359AAE	Hyderabad Shamshabad H (Telangana)	0.0904	120,774	99.5%

Interpretation: The Gurgaon Bilaspur hub is the single largest chokepoint, with 22.02% of all shortest paths in the network passing through it. It exhibits a near-universal 95.8% SLA breach rate, representing the highest priority for operational upgrades.

Recommended Corridor-Specific Interventions

For chronically delayed corridors (where actual time exceeds OSRM by > 20%), we recommend the following strategic shifts:

- 1. **Parallel Route Expansion:**
- 2. *Corridor:* Bangalore Nelamangala to Bengaluru Bommasandra.

3. *Action:* Route traffic through alternative arterial bypasses to avoid Nelamangala's local peak-hour congestion.
4. **Facility Processing Upgrade:**
5. *Corridors involving Gurgaon Bilaspur:* Sonipat Kundli and Bhiwandi Mankoli.
6. *Action:* Introduce automated sortation systems and expand sorting bays to reduce dwell time.
7. **Route-Type Shift (Carting to FTL):**
8. *Corridors like Agra Idgah to Delhi Gateway:*
9. *Action:* Switch route type from Carting to FTL. Historical data shows that FTL is **49.5 minutes faster** on average for this corridor due to fewer intermediate transshipment hops.

Projected Operational & Financial Impact

Our simulation shows that upgrading the top 3 hubs (Gurgaon Bilaspur, Bangalore Nelamangala, and Bhiwandi Mankoli) to resolve processing bottlenecks and reduce delay times by 50% yields: - **1.83% absolute reduction in late deliveries** (preventing hundreds of SLA breaches in our sample alone). - Assuming an average SLA penalty and customer churn cost of **INR 500 per late delivery**, this intervention recovers **INR 94,500 in direct revenue-at-risk** from our training sample alone, scaling to **millions of rupees** annually across our full network volume. - **Improved downstream capacity planning** due to tighter ETA predictions (19.88% lower error).